

Model ↓	Training Cutoff ↓	$\mathcal{F}^{2022}$	$\mathcal{F}_{\text{decay}}^{2022}$	$\mathcal{F}^{2021}$	$\mathcal{F}_{\text{decay}}^{2021}$	$\mathcal{F}_{\text{max}}$
LLaMA1-65B	Aug. 2022	12.1	26.8	13.2	29.5	54.5
GPT-3	Sep. 2021	10.0	22.9	11.4	25.1	46.5
ChatGPT	Sep. 2021	17.1	32.5	23.8	35.5	46.0
Tulu2-70B	-	16.2	32.2	19.1	35.1	50.1
LLaMA2-70B	Sep. 2022	17.2	34.1	19.8	37.0	55.9
+ prompting to 2021		19.7 (+2.5)	38.9 (+4.8)	25.9 (+6.1)	42.4 (+5.4)	56.7 (+0.8)
+ prompting to 2022		27.4 (+10.2)	40.3 (+6.2)	23.9 (+4.1)	40.5 (+3.5)	54.1 (-1.8)
+ finetuning to 2021		20.8 (+3.6)	39.3 (+5.2)	29.2 (+9.4)	42.5 (+5.5)	56.7 (+0.8)
+ finetuning to 2022		27.9 (+10.7)	40.7 (+6.6)	23.7 (+3.9)	40.8 (+3.8)	55.4 (-0.5)
+ adaptive finetuning		25.7 (+8.5)	40.0 (+5.9)	23.5 (+3.7)	40.9 (+3.9)	57.9 (+2.0)

Table 3: Performance of temporally unaligned and aligned model on the TAQA dataset. Here, we mainly target 2021 and 2022 for recency alignment, and we evaluate the aligned LMs based on the target-year ($\mathcal{F}^{2022}$ and $\mathcal{F}^{2021}$), temporal-decayed ($\mathcal{F}_{\text{decay}}^{2022}$ and $\mathcal{F}_{\text{decay}}^{2021}$), and historical-max ($\mathcal{F}_{\text{max}}$) F1 scores, as defined in §2.3.

Data selection ↓	$\mathcal{F}^{2022}$	$\mathcal{F}_{\text{decay}}^{2022}$	$\mathcal{F}_{\text{max}}$
All data	19.8	30.9	44.9
Random	19.5 (-0.3)	30.5 (-0.4)	44.3 (-0.6)
Popularity	19.9 (+0.1)	31.3 (+0.4)	45.1 (+0.2)
Model confidence	19.5 (-0.3)	30.7 (-0.2)	44.9 (+0.0)
Model correctness	20.5 (+0.7)	31.8 (+0.9)	45.5 (+0.6)

Table 4: Training data selection analysis with LLaMA2-13B model for optimizing the temporal alignment.

lection methods select 5000 examples from TAQA training set for fair comparison.

Table 4 compares these methods, proving the superior performance of correctness-based selection. This additionally confirms that the finetuning is activating models’ internal knowledge, rather than injecting new knowledge into the model. Otherwise, the popularity-based or full-data training will lead to better performance.

4.4 Improvement Analysis

We provide more detailed analyses of how temporal alignment improves model predictions and confirm that the improvement is not from trivial factors.

Popular knowledge gets aligned better. We first illustrate the association between question popularity and models’ improvement on them through temporal alignment, shown in Figure 4 (left). Overall, the target-year F1 scores of both unaligned and aligned models improve as the popularity grows, confirming that popular questions are easier for LMs to answer. Moreover, the improvement from temporal alignment also increases on popular questions, indicating that models have more potential to be aligned on popular topics, since such knowledge is better memorized during pretraining but not triggered due to the temporal chaos.

Improvement is not from memorizing facts in finetuning. The surprising effectiveness of doing temporal alignment through finetuning might be weakened if it is mainly due to some knowledge overlap between the finetuning and testing sets. To confirm this is not the reason, Figure 4 (right) plots the relationship between testing questions’ semantic similarity⁶ to the training set and models’ performance on them. We see that the improvement from alignment does not increase as the similarity grows. This implies that memorization is not the reason for better performance on TAQA.

Models are aligned beyond just QA formatting.

We further test whether the temporally aligned LMs are learning to activate recent knowledge rather than just learning to answer questions in the correct format. We conduct another two finetunings, with 1) data randomly sampled from NaturalQuestions (NQ, Kwiatkowski et al., 2019) and 2) data randomly sampled from TAQA paired with answers picked from random years. We see from Figure 5 that both of them generally do not have the effect of aligning the model to a recent time. Interestingly, NQ actually aligns the model towards 2017, which we speculate is because the dataset was constructed using much information back then.

4.5 Scaling Trend w.r.t. Model Size

We compare the performance of LLaMa2 70B, 13B, and 7B models before and after finetuning to analyze the effect of model size on temporal alignment. As is shown in Table 5, the improvement of finetuned 7B and 13B models on target-year F1 score ($\mathcal{F}^{2022}$) are both lower than that of the 70B model.

⁶We use sentence-BERT embeddings (Reimers and Gurevych, 2019) to compute the similarity.

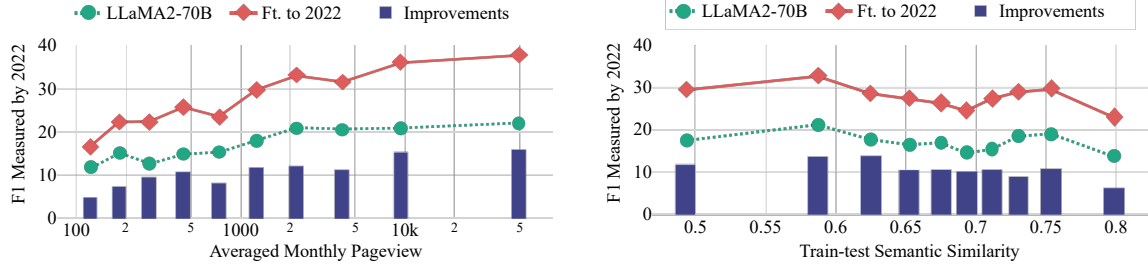


Figure 4: (Left) Relationship between question popularity (measured by the pageviews of their originated Wikipedia page) and models' F1 score on them, as in 2022. (Right) Relationship between testing questions' maximum semantic similarity to the training set and models' F1 score on them, as in 2022.

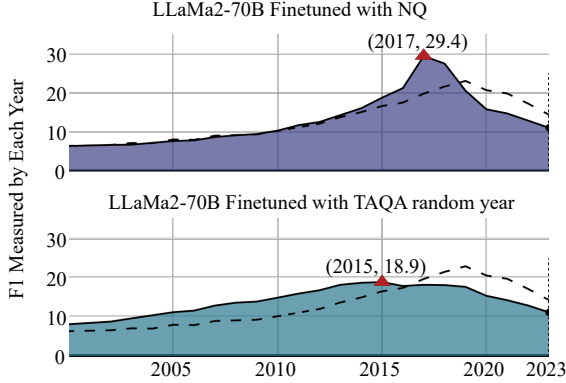


Figure 5: The temporal knowledge distribution of LMs finetuned with the NQ dataset and our TAQA dataset where the answers are randomly sampled.

Method	Metric	7B	13B	70B
Unaligned	$\mathcal{F}^{2022}$	10.4	13.0	17.2
	$\mathcal{F}_{\text{decay}}^{2022}$	21.6	25.6	34.1
	$\mathcal{F}_{\text{max}}$	41.1	45.5	55.9
Finetuned	$\mathcal{F}^{2022}$	17.6 (+7.2)	20.5 (+7.5)	27.9 (+10.7)
	$\mathcal{F}_{\text{decay}}^{2022}$	28.5 (+6.9)	31.8 (+6.2)	40.7 (+6.6)
	$\mathcal{F}_{\text{max}}$	43.1 (+2.0)	46.5 (+1.0)	55.4 (-0.5)
Adaptive	$\mathcal{F}^{2022}$	17.3 (+6.9)	18.9 (+5.9)	25.7 (+8.5)
	$\mathcal{F}_{\text{decay}}^{2022}$	28.8 (+7.2)	32.0 (+6.4)	40.0 (+5.9)
	$\mathcal{F}_{\text{max}}$	44.9 (+3.8)	50.2 (+4.7)	57.9 (+2.0)

Table 5: Effect of LLaMA2 model sizes on temporal alignment, when finetuning to the year 2022.

Moreover, considering the historical-max F1 score ($\mathcal{F}^{\text{max}}$), the 70B model sees diminishing improvement after temporal alignment finetuning, while 7B and 13B models keep benefiting from finetuning. We speculate that the reason is the 70B pretrained model memorizes more factual knowledge, which might suffer from more forgetting.

5 Further Analysis: To What Year Are LMs Most Readily Alignable?

Aside from aligning LMs to a recent time, we are also curious about which year, historically speaking, an LM is most readily alignable to and achieve

the best performance. We investigate this by finetuning LLaMa-70B to different years. Figure 6 shows that 2019 is the best year for aligning the LLaMa2-70B model instead of the year of the pre-training cutoff. Meanwhile, the performance of models finetuned with data whose answers are true in 2010, 2015, and 2019 are all better than in recent years (2022 and 2023). We believe this is because the information about the most recent factual knowledge is too short in existing pretraining corpora, so it is more challenging to align the internal knowledge of LMs to recent times than slightly further to the past.

Furthermore, if considering the historical maximum $\mathcal{F}_{\text{max}}$, the target year of 2015 performs the best among all our target year choices. We argue that the reason could be factual knowledge in 2015 aligns the most with LLaMa2-70B's internal (but unactivated) knowledge learned from pretraining, and therefore, finetuning the LM on TAQA where the answers were true in 2015 causes the least contradiction and thus makes the LM the least likely to produce historically-incorrect answers.

6 Related Work

Temporal Misalignment in LMs Temporal misalignment is a much-studied issue for LMs. For instance, Lazaridou et al. (2021) pointed out that LMs suffer from the temporal gap between their training and deployment data, and Jang et al. (2022) proposed the benchmark of training and evaluating LMs throughout time. Luu et al. (2022) showed that temporal misalignment also hurts LMs' performance on downstream finetuning tasks, and following research indicated this effect can be task-dependent (Agarwal and Nenkova, 2022). Recent research suggested that chaotic pretraining corpus is one of the reasons for LMs' temporal misalignment (Longpre et al., 2023), and LMs can represent temporal knowledge learned from pretraining in

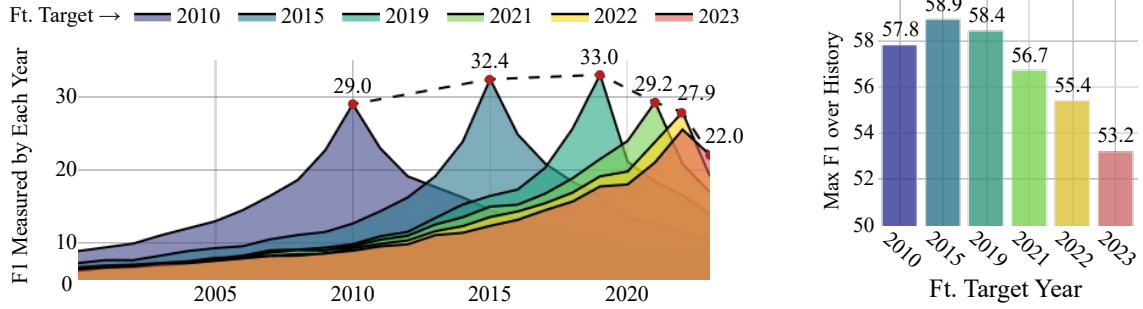


Figure 6: LMs’ F1 score in every year between 2000-2023 (left) along with their historical-max F1 scores $\mathcal{F}^{\max}$ (right) after being finetuned to 2010, 2015, 2019, 2021, 2022, and 2023. The dotted line shows the trend of each aligning year’s F1 score evaluated with that year’s answers.

their internal states (Gurnee and Tegmark, 2023). These findings open up the possibility of aligning models to a specific time. Different techniques have been proposed to tackle the temporal misalignment issue (Zhang et al., 2023), including continual pretraining (Jin et al., 2022; Loureiro et al., 2022; Ke et al., 2023), knowledge editing (Mitchell et al., 2022; Meng et al., 2023), and retrieval augmentation (Zhang and Choi, 2023; Vu et al., 2023). These methods edit LMs’ knowledge instead of eliciting the knowledge already learned from pretraining, as we aim to do in this work.

Temporal QA and Reasoning Temporal QA and reasoning are tasks where the correct answers change through time (Figuerola, 2010), e.g., “Who is the president of the United States”. Prior research tried using knowledge graphs to tackle these tasks (Jia et al., 2021; Saxena et al., 2021; Shang et al., 2022). As modern LMs can memorize factual knowledge, new benchmarks (Chen et al., 2021; Dhingra et al., 2022a; Tan et al., 2023; Wei et al., 2023) address the challenge of leveraging LMs to answer questions with contextual temporal mentions provided (e.g., “in 2023”, “after” certain events, etc.). However, such question structure can sound unnatural, and these studies do not explicitly focus on questions whose answers have changed in recent years. Other news article-based datasets like RealtimeQA (Kasai et al., 2022) and StreamingQA (Liska et al., 2022), despite having more natural questions, still lack questions whose answers can change multiple times, rather than a single time. In contrast, our TAQA dataset curates natural questions without temporal information but with at least five different answers after 2000. Several approaches have been proposed to tackle the temporal QA and reasoning challenges (Son and Oh, 2023; Tan et al., 2023), yet these approaches re-

quire dedicated training for enhancing LMs’ understanding of temporal mentions, which are missing in most natural questions.

7 Conclusion

We build TAQA, a QA dataset with time-sensitive questions during a recent time period (2020–2023). Using this dataset, we quantitatively show that LMs can respond to questions chaotically in terms of the question’s placement in time. We thus propose the concept of temporal alignment and a set of methods for it. Extensive experiments demonstrate the possibility and effectiveness of aligning model internal knowledge temporally, both to a recent time and to a historical time.

8 Limitations

Throughout the development of this project, we identified several limitations that invite additional investigation. Firstly, in our dataset construction, we use GPT-4 to generate natural language questions from tabular Wikipedia data without human annotation. While we do adopt quality control policies, it may still be the case that there are noisy generations. These noisy generations, where the question’s text content might not be consistent with the quality of a human-written question, may impact the usefulness of such data in the alignment process and thus the effectiveness overall. Secondly, it is difficult to determine the exact date and time that knowledge changes. This, coupled with the fact the exact pretraining corpora of LLaMa2 and similar models are unknown, means we cannot determine whether an LM has observed certain knowledge. This can introduce inaccuracies to scoring when measuring both the original year the model is internally aligned to, and the year it is aligned to after our methods are applied. Thirdly,